CRYSS



Crystallization Screening System

Presentation 2

www.cryss.co.uk

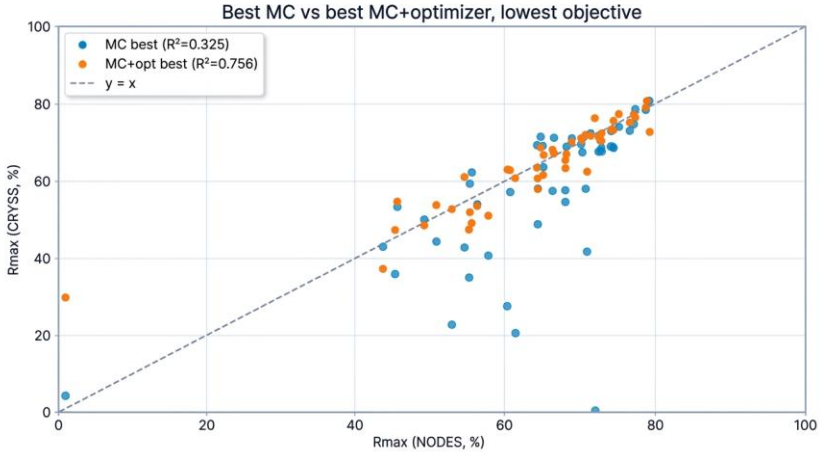
4-Nov-2026 (updated 9-Nov-2026)

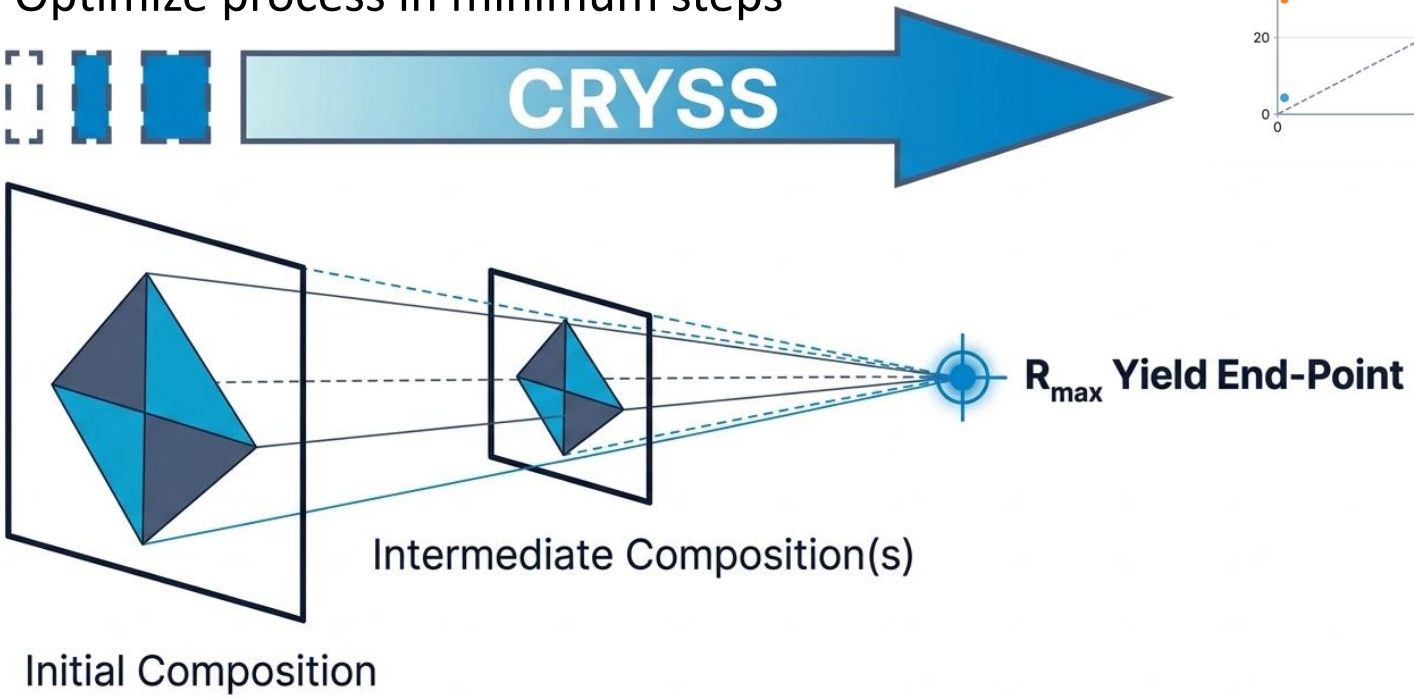
The CRYSS Approach



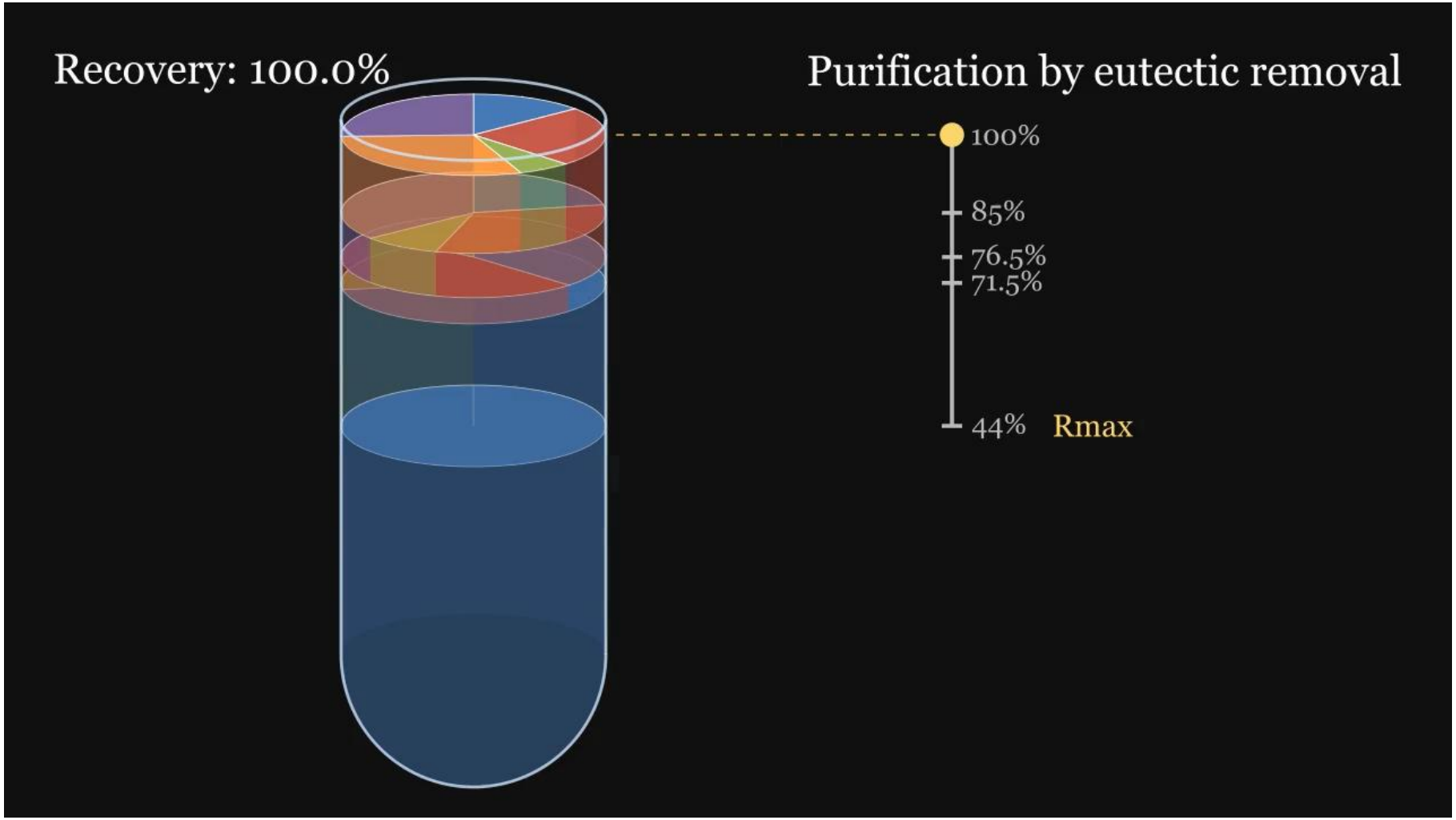
- Mechanistic model & fitted to experimental data
- Only requires data from partial purifications
  - Rank systems
  - Predict solubility diagram
  - Optimize process in minimum steps

Single solvent volume composition slice



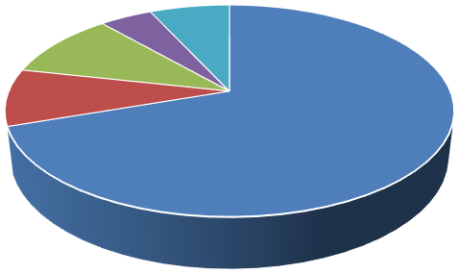


Theory: Purification by Stages



https://www.manim.community/

Starting Composition (Xc)



Strategy to Validate the Approach



- Use published equilibrium partitioned data in a 4-component system comprising amino indanol stereoisomers with salt-formers:
  - Some systems were close to the Ideal Approximation and some deviated significantly
  - We are focussing on (observed) real purification behaviour and we know that CRYSS absorbs deviations from ideality
  - We use CRYSS V2.0 visualization tools to explore the data
- Objective:
  - To demonstrate real behaviour matching the theoretical “Purification by Stages” as predicted in *Tet. Asym*., **1998**, 9, 2925; nb. we know that solving multicomponent eutectics was established in the field of liquid crystals at Hughes Research Laboratories (Mol. Cryst. Liq. Cryst., 1981, 76, 211)
  - Underpinning NODES & CRYSS, on which their technology is based

Reference Experimental Data



Prediction of Maximum Yield in the Crystallization of Multicomponent Isomeric Systems

Alan A. Smith,*,† Ben McKay, Eric W. P. Damen,‡ Simone Darphorn-Hooijschuur, Erik-Jan Ras, and Goran Verspui‡
Avantium Technologies BV, Zekeringstraat 29 1014 BV, The Netherlands

Abstract:
We have developed a computer program (CRYSS), which may be used to rank the performance of a series of crystallization experiments. The method, at its simplest, allows the chemist to take composition data from partially purified crystallized solids and mother liquors; when this is coupled with initial composition data, the program can predict the maximum yield the system is capable of delivering after optimization. In this paper the underlying theoretical basis of the method is investigated by means of the crystallization of a three-component blend of hydrobenzoin isomers, along with the investigation of a four-component system based on aminoindanol. In the latter case, the modulation of purification performance via salt formation is demonstrated.

yields. Moreover, ranking is possible when partial purification has been obtained, in principle, from a single experiment. The method involves using a numerical optimizer to fit parameters to a mechanistic model; the input data at its simplest is (i) composition data from the partially purified crystallized solid, (ii) mother liquor compositional analysis, and (iii) the starting composition.

Conglomerate Enantiomer Mixtures. Consider the simple case of a pair of enantiomers behaving as a conglomerate. These have identical melting points and heats of fusion and the same molecular architecture, differing only in that they are mirror images. A eutectic exists and is of a 50:50 composition. This is true, irrespective of the choice of solvent. This fact makes it relatively straightforward to purify an enantiomerically enriched mixture (i.e. one which may result from an asymmetric synthesis). To illustrate, let

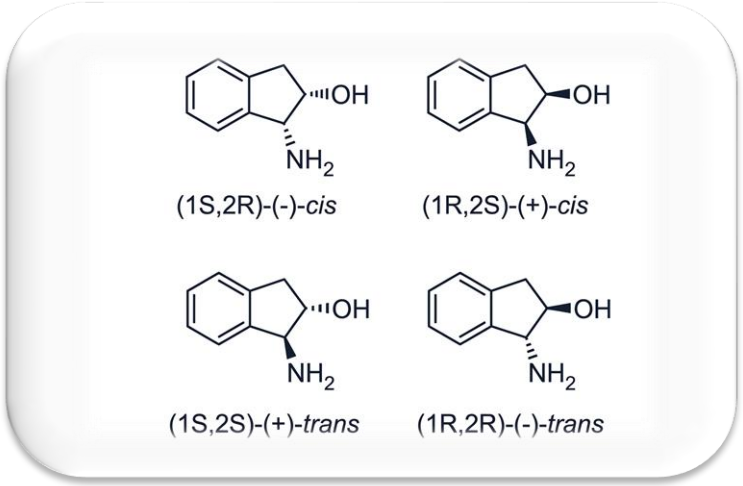


Table 3. **Selection of Table 12 where CRYSS R_{max} correlates well with ideal R_{max}**

		initial composition (%)				solid composition (%)				CRYSS	recovery (R_{obs})(%)	MultXeu
expt #	experiment	1S,2R	1R,2S	1R,2R	1S,2S	1S,2R	1R,2S	1R,2R	1S,2S	R_{max} (%)	CRYSS	R_{max} (%)
1	*cis*-tartrate	93.5	2.2	2.2	2.0	97.9	0.5	0.6	1.1	6	17	2
2	*trans*-tartrate	1.5	1.3	1.7	95.4	1.2	1.1	1.9	95.8	2	89	29
3	*cis*-tosylate	94.6	1.7	2.1	1.7	99.3	0.0	0.1	0.6	67	76	80
4	*trans*-tosylate	1.5	2.7	1.9	94.0	0.1	0.1	0.0	99.8	60	68	84
5	*cis*-salicylate	94.3	1.7	1.9	2.1	32.3	0.0	0.0	67.7	1	1	0
8	*trans*-bromide	1.6	2.7	1.6	94.1	0.0	0.0	0.0	99.9	72	73	93
10	*trans*-formate	1.9	1.5	1.3	95.3	0.1	1.1	0.6	98.2	71	85	91
15	*cis*-phenylacetate	89.6	4.2	3.7	2.5	50.3	0.0	19.8	30.0	0	1	2
16	*trans*-phenylacetate	1.6	3.1	4.7	90.5	0.0	0.1	0.1	99.8	79	79	80

Table 4. **Selection of Table 12 where CRYSS R_{max} correlates moderately with ideal R_{max}**

		initial composition (%)				solid composition (%)				R_{max} (%)	recovery (R_{obs})(%)	MultXeu
expt #	experiment	1S,2R	1R,2S	1R,2R	1S,2S	1S,2R	1R,2S	1R,2R	1S,2S	CRYSS	CRYSS	R_{max} (%)
9	*cis*-formate	94.7	1.5	1.7	2.1	99.0	0.0	0.2	0.8	49	61	26
12	*trans*-benzoate	1.7	1.7	94.3	2.4	0.7	0.3	97.5	1.5	45	73	86
17	*cis*-mesylate	1.8	95.0	1.6	1.6	0.0	98.9	0.8	0.3	61	81	90

Table 5. **Selection of Table 12 where CRYSS R_{max} correlates poorly with ideal R_{max}**

		initial composition (%)				solid composition (%)				R_{max} (%)	recovery (R_{obs})(%)	MultXeu
expt #	experiment	1S,2R	1R,2S	1R,2R	1S,2S	1S,2R	1R,2S	1R,2R	1S,2S	CRYSS	CRYSS	R_{max} (%)
6	*trans*-salicylate	2.2	2.0	2.0	93.8	37.5	0.0	0.0	62.5	1	2	72
7	*cis*-bromide	94.6	1.6	1.6	2.3	91.9	0.1	3.4	4.6	1	46	86
11	*cis*-benzoate	2.2	92.9	2.5	2.5	3.0	90.6	3.5	2.9	2	70	74
18	*trans*-mesylate	1.9	1.9	94.1	2.0	0.1	2.4	97.5	0.1	2	77	90

Org. Process Res. Dev. **2006**; 10 (6): 1132–1143 (see also supporting data zip file with publication)

OPRD Paper – Supporting Info



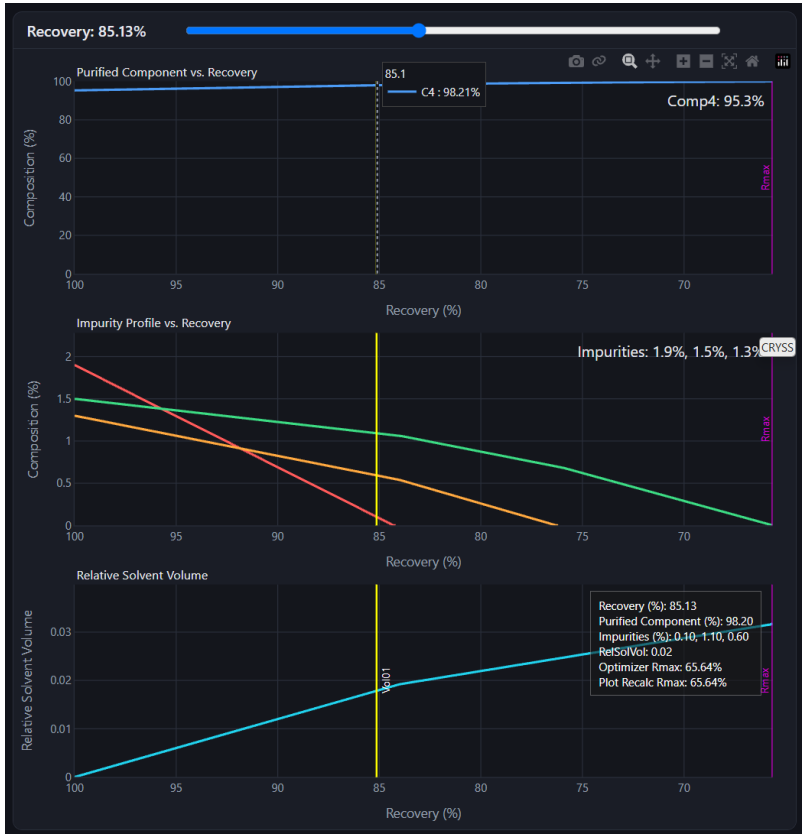
This is the data was used in the CRYSS Calculations located in “MultXeu - Ideal Approx Calcs using Final DSC Data and Final Initial Compositions.XLS”

Download available from ACS at:- https://ndownloader.figstatic.com/files/4752718

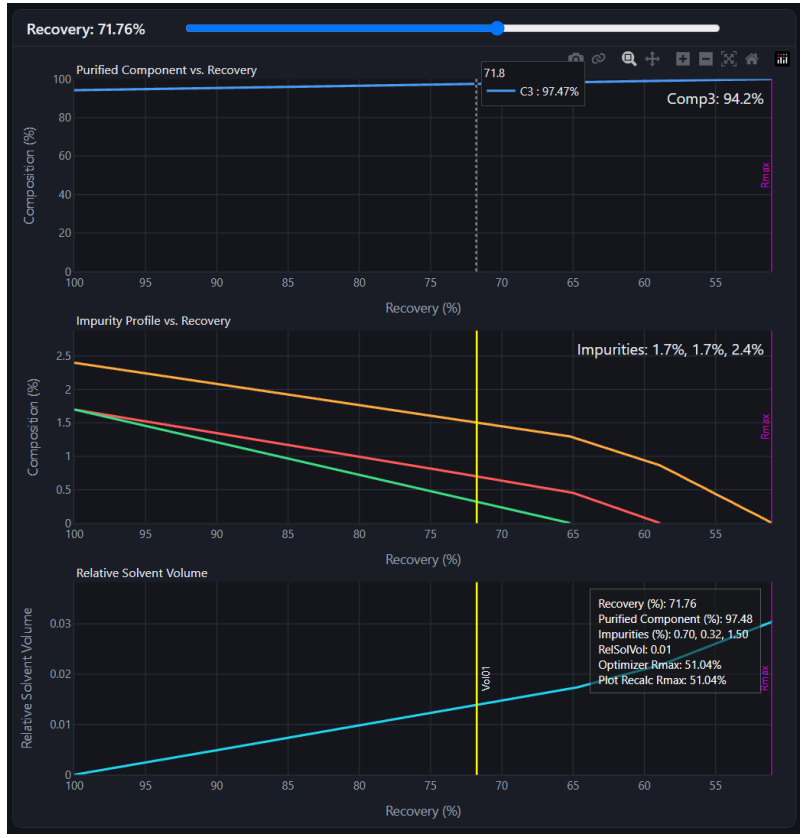
Expt #	Salt Type	Initial Composition				Liquor Composition				Solid Composition				CRYSS	Obs	Ideal
	(major isomer)	1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	R_{max}	Rec.	R_{max}
1 S L	tartrate (cis)	93.50%	2.20%	2.20%	2.00%	92.70%	2.60%	2.60%	2.20%	97.90%	0.50%	0.60%	1.10%	6%	17%	2%
2 S L	tartrate (trans)	1.50%	1.30%	1.70%	95.40%	4.10%	3.30%	0.00%	92.60%	1.20%	1.10%	1.90%	95.80%	2%	89%	29%
3 S L	tosylate (cis)	94.60%	1.70%	2.10%	1.70%	79.50%	7.00%	8.40%	5.10%	99.30%	0.00%	0.10%	0.60%	67%	76%	80%
4 S L	tosylate (trans)	1.50%	2.70%	1.90%	94.00%	4.40%	8.10%	5.80%	81.70%	0.10%	0.10%	0.00%	99.80%	60%	68%	84%
5 S L	salicylate (cis)	94.30%	1.70%	1.90%	2.10%	95.10%	1.70%	1.90%	1.20%	32.30%	0.00%	0.00%	67.70%	1%	1%	0%
6 S L	salicylate (trans)	2.20%	2.00%	2.00%	93.80%	1.60%	2.00%	2.00%	94.40%	37.50%	0.00%	0.00%	62.50%	1%	2%	72%
7 S L	bromide (cis)	94.60%	1.60%	1.60%	2.30%	96.80%	2.90%	0.00%	0.30%	91.90%	0.10%	3.40%	4.60%	1%	46%	86%
8 S L	bromide (trans)	1.60%	2.70%	1.60%	94.10%	5.90%	9.60%	6.00%	78.50%	0.00%	0.00%	0.00%	99.90%	72%	73%	93%
9 S L	formate (cis)	94.70%	1.50%	1.70%	2.10%	87.80%	3.90%	4.10%	4.20%	99.00%	0.00%	0.20%	0.80%	49%	61%	21%
10 S L	formate (trans)	1.90%	1.50%	1.30%	95.30%	12.30%	3.60%	5.30%	78.80%	0.10%	1.10%	0.60%	98.20%	71%	85%	91%
11 S L	benzoate (cis)	2.20%	92.90%	2.50%	2.50%	0.30%	98.20%	0.10%	1.40%	3.00%	90.60%	3.50%	2.90%	2%	70%	74%
12 S L	benzoate (trans)	1.70%	1.70%	94.30%	2.40%	4.20%	5.30%	85.80%	4.60%	0.70%	0.30%	97.50%	1.50%	45%	73%	86%
15 S L	phenylacetate (cis)	89.60%	4.20%	3.70%	2.50%	89.90%	4.30%	3.50%	2.30%	50.30%	0.00%	19.80%	30.00%	0%	1%	2%
16 S L	phenylacetate (trans)	1.60%	3.10%	4.70%	90.50%	7.70%	14.40%	22.20%	55.70%	0.00%	0.10%	0.10%	99.80%	79%	79%	80%
17 S L	mesylate (cis)	1.80%	95.00%	1.60%	1.60%	9.40%	78.60%	4.90%	7.10%	0.00%	98.90%	0.80%	0.30%	61%	81%	90%
18 S L	mesylate (trans)	1.90%	1.90%	94.10%	2.00%	8.30%	0.30%	82.60%	8.80%	0.10%	2.40%	97.50%	0.10%	2%	77%	90%

3 of 3 Impurities at Recovery Point

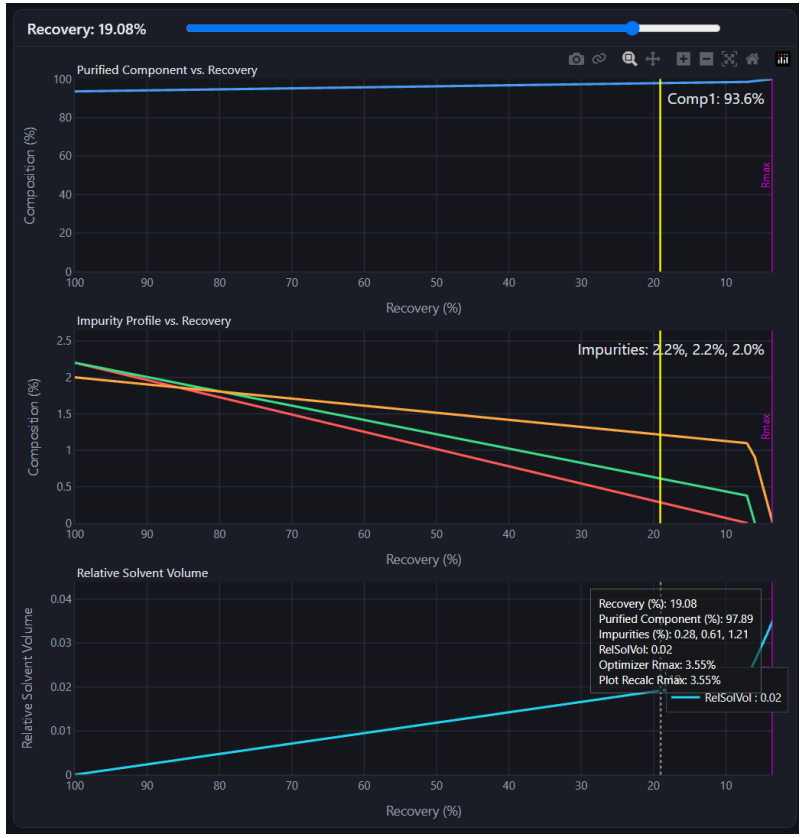
(*trans*)-formate



(*trans*)-benzoate



(*cis*)-tartrate



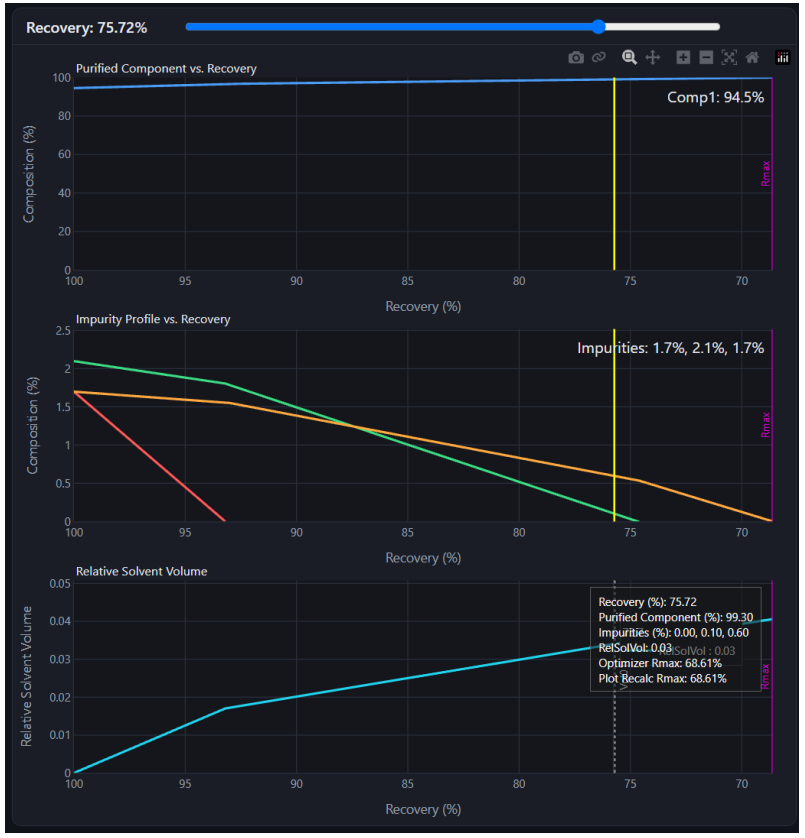
Expt #	Salt Type	Initial Composition (%)			Liquor Composition (%)				Solid Composition (%)				CRYSS (2006)	Observed Recovery (%)
		1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	R_{max}	(R_{obs})
10	(*trans*)-formate	1.9%	1.5%	1.3%	95.3%	12.3%	3.6%	5.3%	78.8%	0.1%	1.1%	0.6%	98.2%	71%	85%
12	(*trans*)-benzoate	1.7%	1.7%	94.3%	2.4%	4.2%	5.3%	85.8%	4.6%	0.7%	0.3%	97.5%	1.5%	45%	73%
1	(*cis*)-tartrate	93.5%	2.2%	2.2%	2.0%	92.7%	2.6%	2.6%	2.2%	97.9%	0.5%	0.6%	1.1%	6%	17%

2 of 3 Impurities at Recovery Point

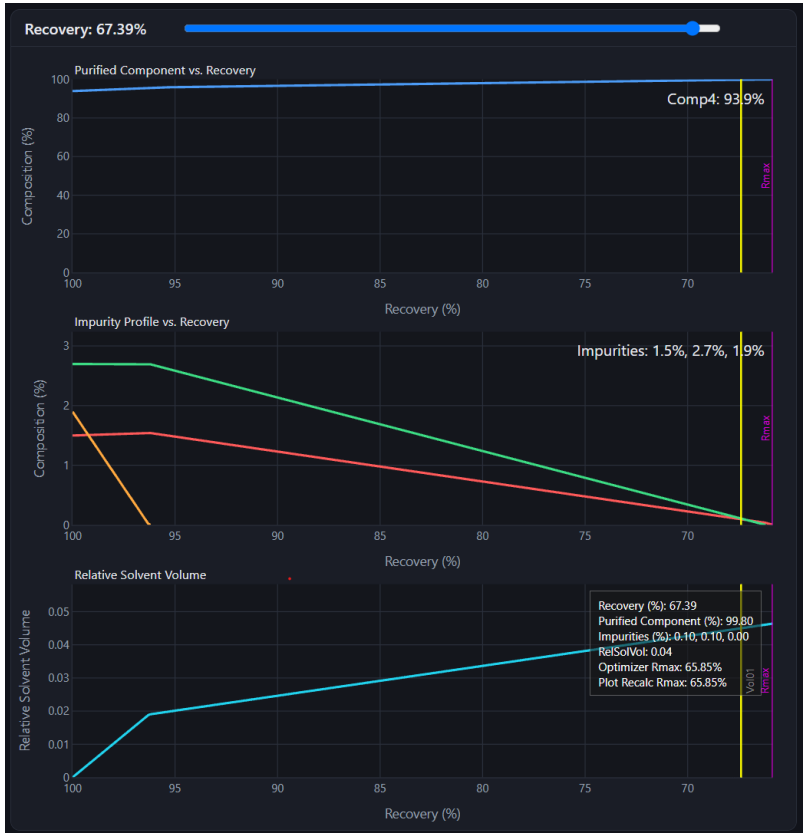
(*cis*)-mesylate*



(*cis*)-tosylate



(*trans*)-tosylate



Expt #	Salt Type	Initial Composition (%)				Liquor Composition (%)				Solid Composition (%)				CRYSS (2006)	Observed Recovery (%)
		1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	R_{max}	(R_{obs})
17	(*cis*)-mesylate	1.8%	95.0%	1.6%	1.6%	9.4%	78.6%	4.9%	7.1%	0.0%	98.9%	0.8%	0.3%	61%	81%
3	(*cis*)-tosylate	94.6%	1.7%	2.1%	1.7%	79.5%	7.0%	8.4%	5.1%	99.3%	0.0%	0.1%	0.6%	67%	76%
4	(*trans*)-tosylate	1.5%	2.7%	1.9%	94.0%	4.4%	8.1%	5.8%	81.7%	0.1%	0.1%	0.0%	99.8%	60%	68%

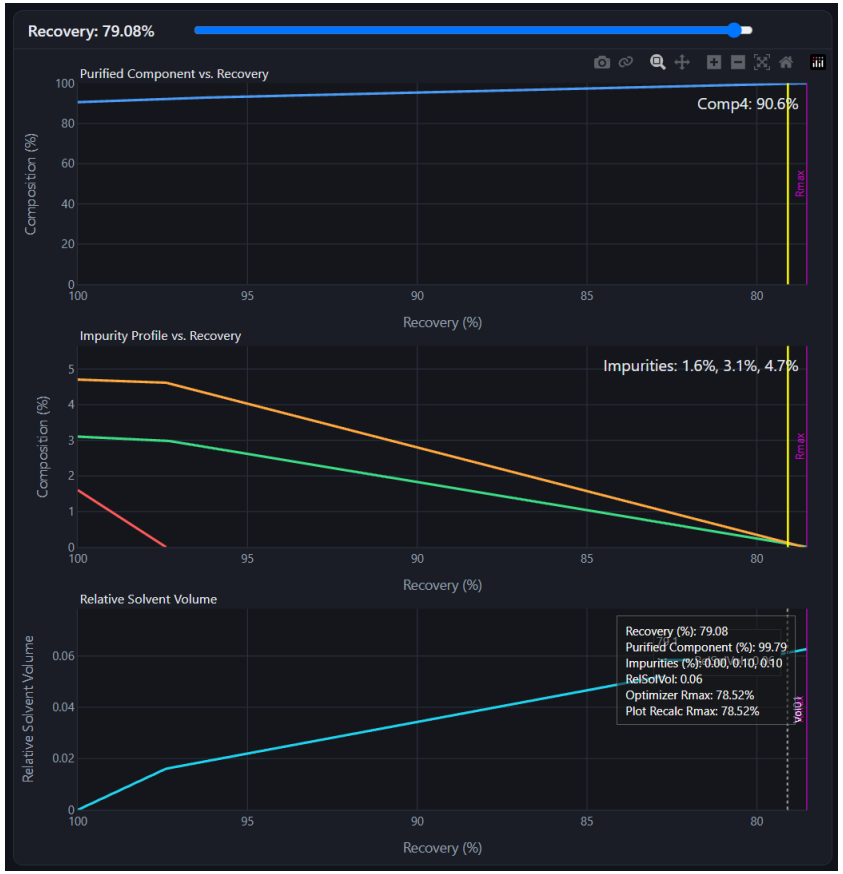
* nb the (cis)-mesylate plot in the online presentation; replotting resulted in a different solution for component 1 solution, as mentioned since it is an unconstrained component, the solution space in the optimizer is infinite

2 of 3 Impurities at Recovery Point

(*cis*)-formate



(*trans*)-phenylacetate

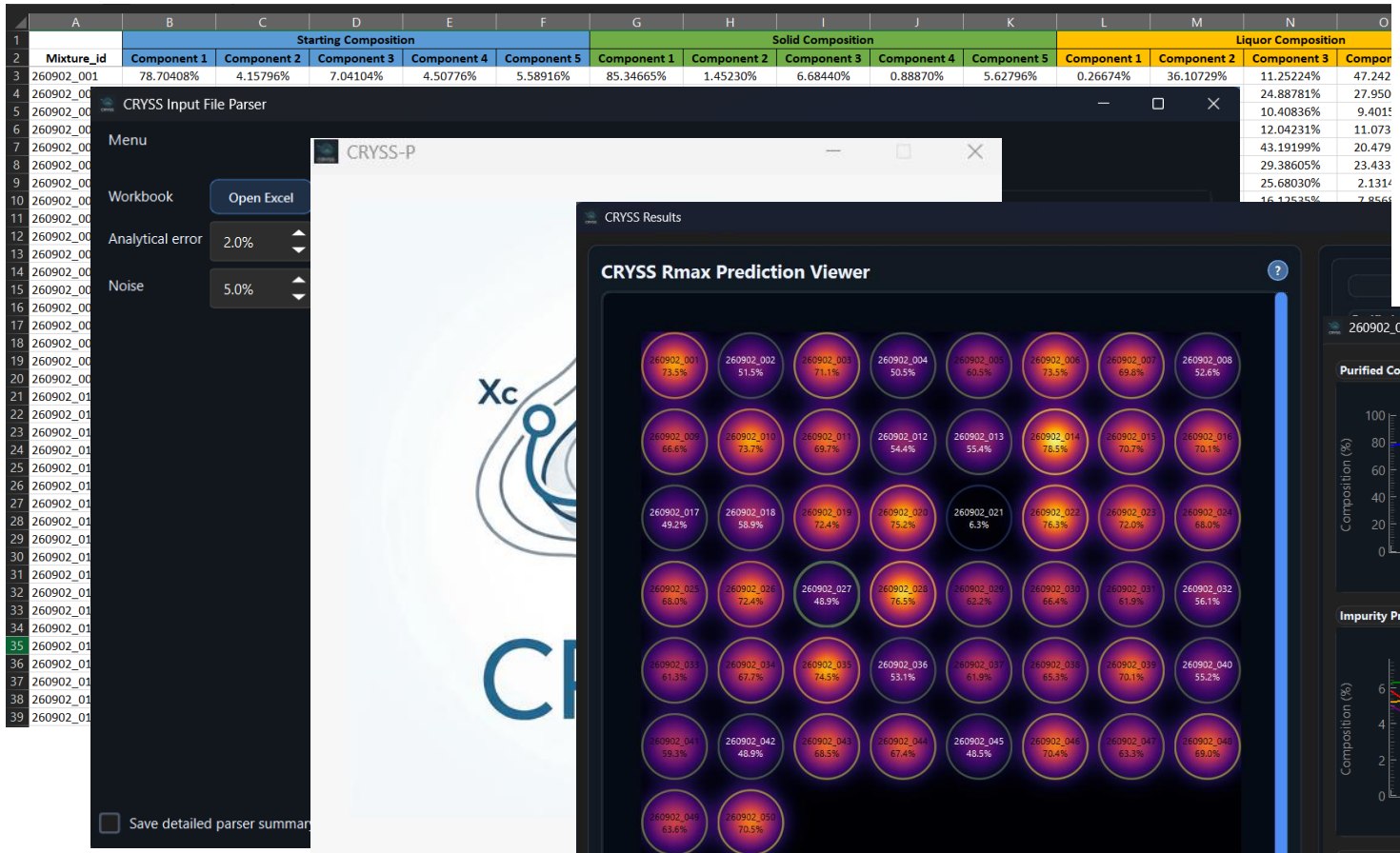


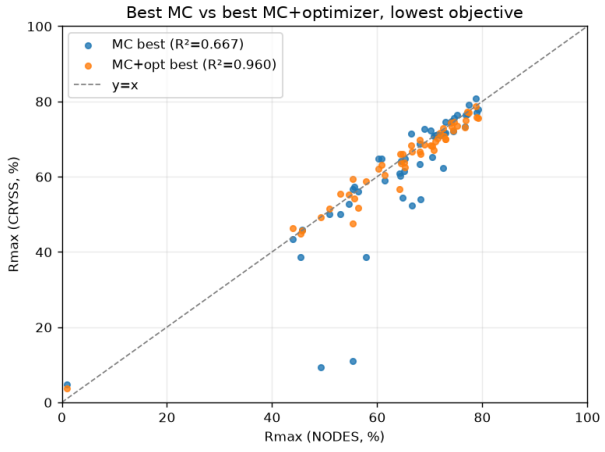
Expt #	Salt Type	Initial Composition (%)				Liquor Composition (%)				Solid Composition (%)				CRYSS (2006)	Observed Recovery (%)
		1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	R_{max}	(R_{obs})
9	(*cis*)-formate	94.7%	1.5%	1.7%	2.1%	87.8%	3.9%	4.1%	4.2%	99.0%	0.0%	0.2%	0.8%	49%	61%
16	(*trans*)-phenylacetate	1.6%	3.1%	4.7%	90.5%	7.7%	14.4%	22.2%	55.7%	0.0%	0.1%	0.1%	99.8%	79%	79%

Anomalous Purification Behaviour

Recalcitrant impurities with devastating impact on yield

Recap: CRYSS Workflow

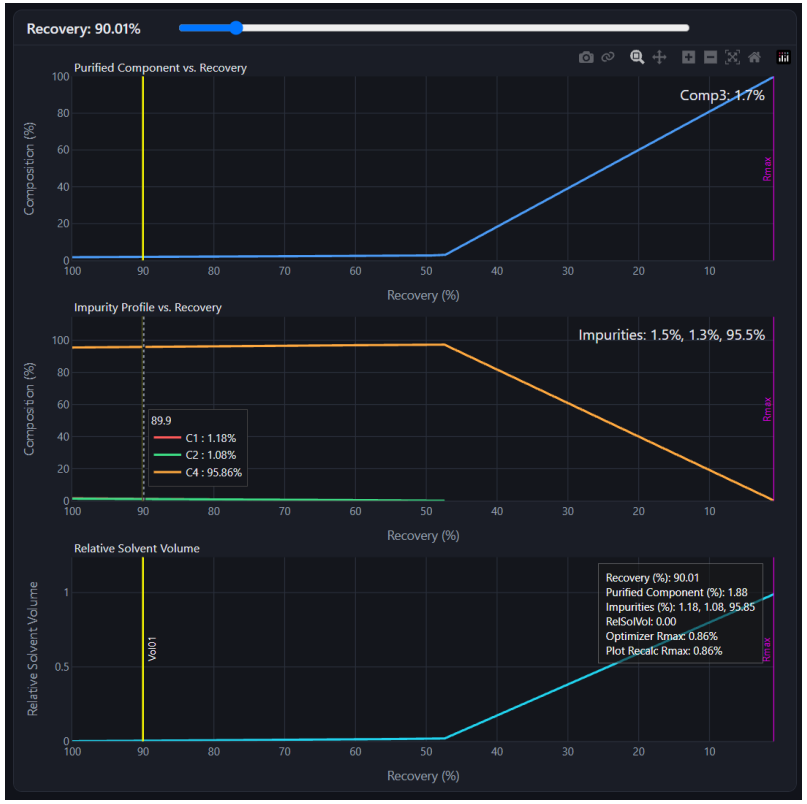






Predicting Minor Component Purifies

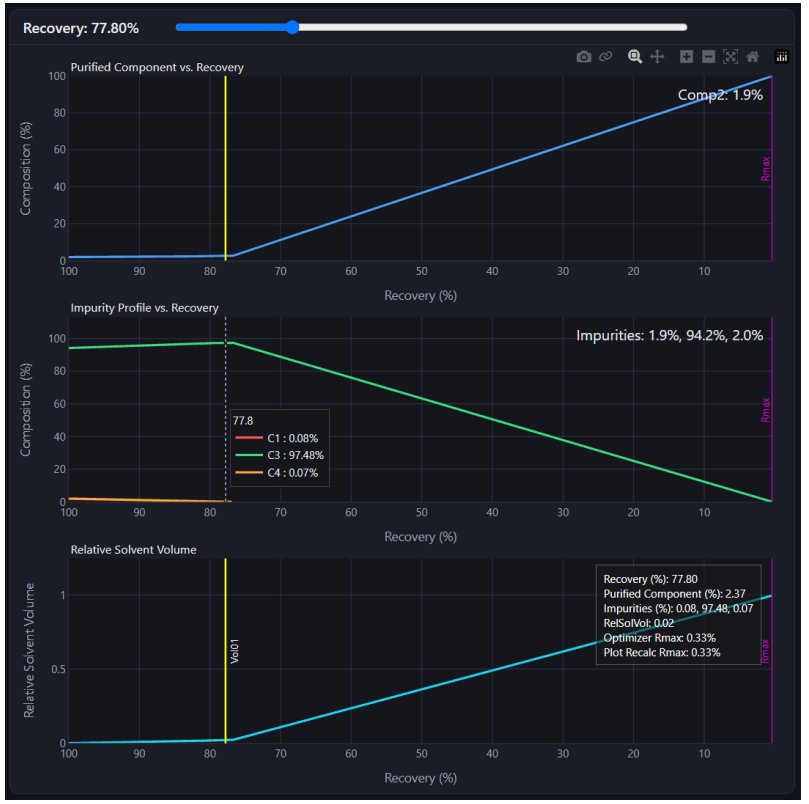
(*trans*)-tartrate



(*cis*)-benzoate



(*trans*)-mesylate*



Expt #	Salt Type	Initial Composition (%)				Liquor Composition (%)				Solid Composition (%)				CRYSS (2006)	Observed Recovery (%)
		1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	R_{max}	(R_{obs})
2	(*trans*)-tartrate	1.5%	1.3%	1.7%	95.4%	4.1%	3.3%	0.0%	92.6%	1.2%	1.1%	1.9%	95.8%	2%	89%
11	(*cis*)-benzoate	2.2%	92.9%	2.5%	2.5%	0.3%	98.2%	0.1%	1.4%	3.0%	90.6%	3.5%	2.9%	2%	70%
18	(*trans*)-mesylate	1.9%	1.9%	94.1%	2.0%	8.3%	0.3%	82.6%	8.8%	0.1%	2.4%	97.5%	0.1%	2%	77%

* nb the (*trans*)-mesylate plot in the online presentation differs; it was recognized that the V2.0 optimizer dropped one of the low-level impurity at recovery point; code has been revised and thus corrected in this plot

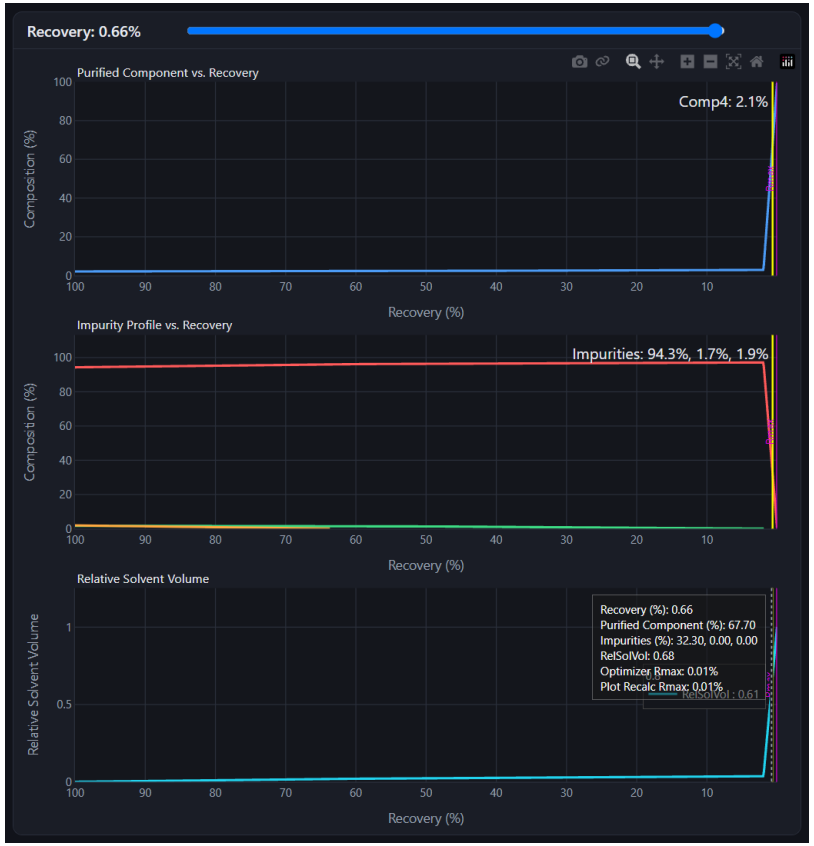
Predicting Minor Component Purifies



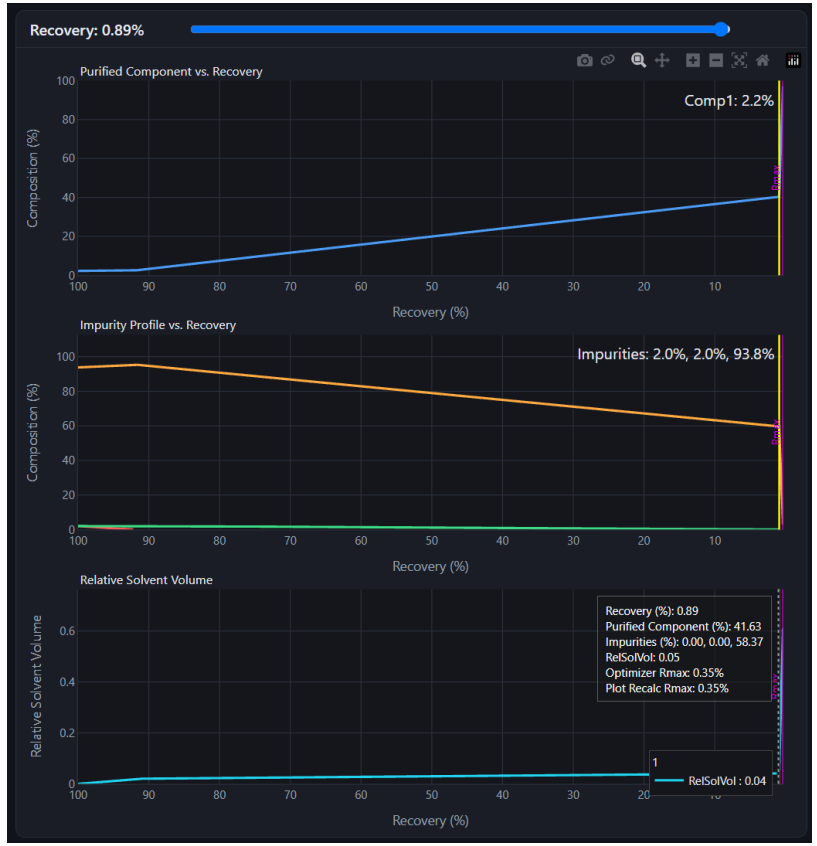
(*cis*)-bromide



(*cis*)-salicylate

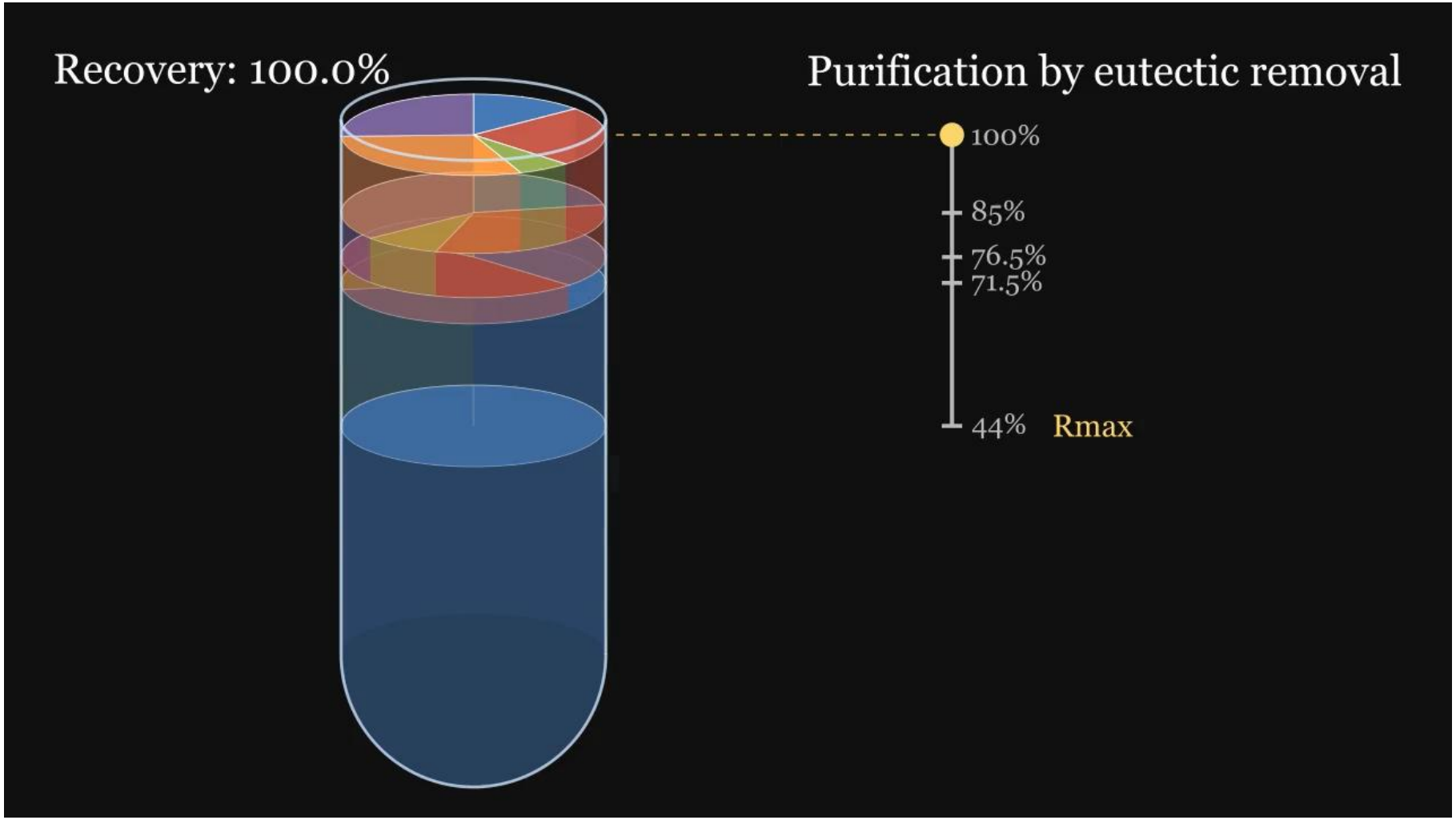


(*trans*)-salicylate



Expt #	Salt Type	Initial Composition (%)				Liquor Composition (%)				Solid Composition (%)				CRYSS (2006)	Observed Recovery (%)
		1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	1S, 2R	1R, 2S	1R, 2R	1S, 2S	R_{max}	(R_{obs})
7	(*cis*)-bromide	94.6%	1.6%	1.6%	2.3%	96.8%	2.9%	0.0%	0.3%	91.9%	0.1%	3.4%	4.6%	1%	46%
5	(*cis*)-salicylate	94.3%	1.7%	1.9%	2.1%	95.1%	1.7%	1.9%	1.2%	32.3%	0.0%	0.0%	67.7%	1%	1%
6	(*trans*)-salicylate	2.2%	2.0%	2.0%	93.8%	1.6%	2.0%	2.0%	94.4%	37.5%	0.0%	0.0%	62.5%	1%	2%

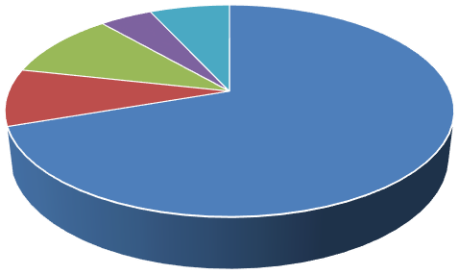
Theory: Purification by Stages



https://www.manim.community/

CRYSS

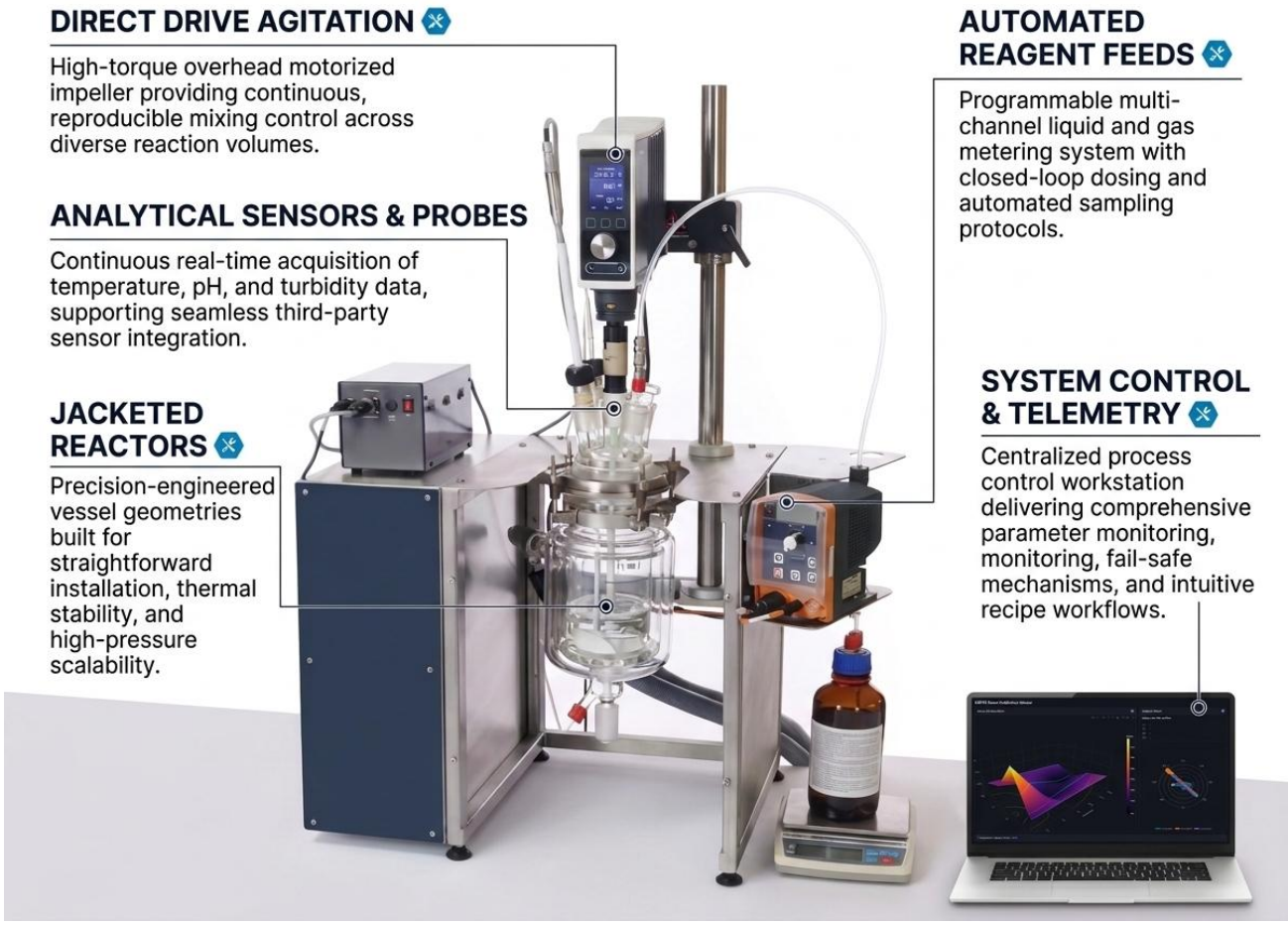
Starting Composition (Xc)



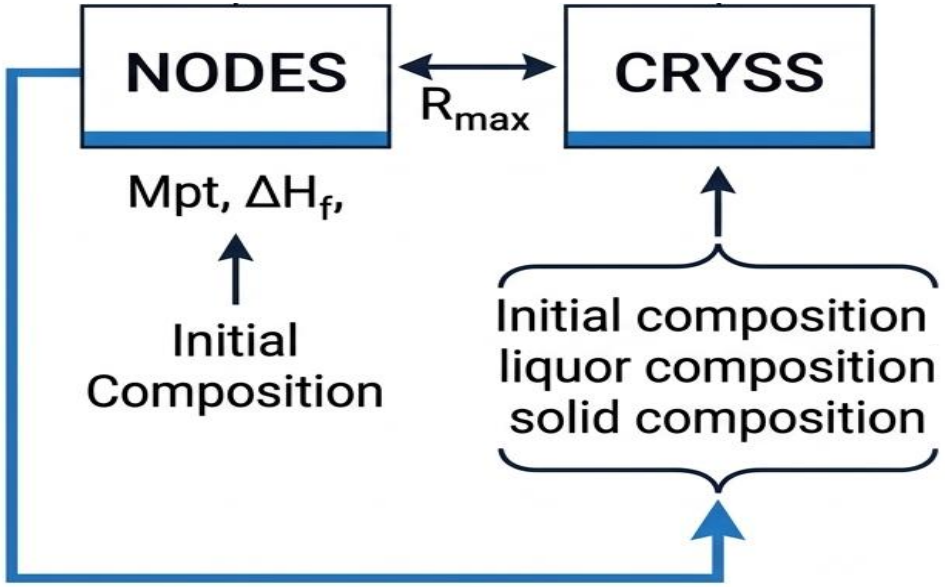
CRYSS

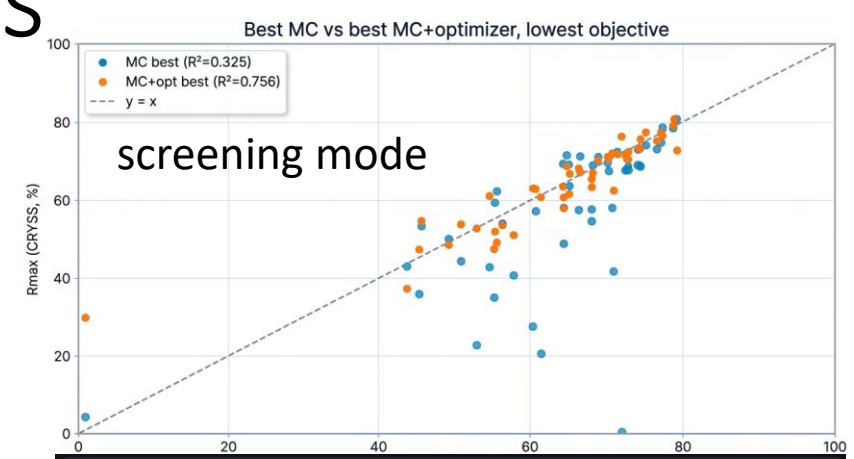
A New Horizon

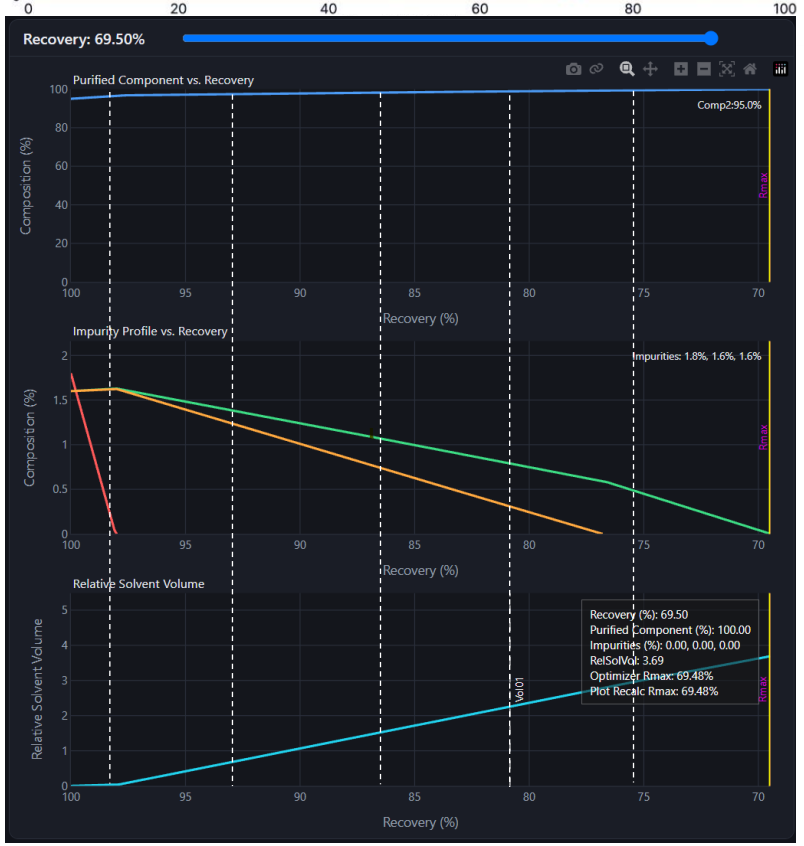
CRYSS-T a powerful process optimization method for CRYSS V3.0



High Resolution Data Sets





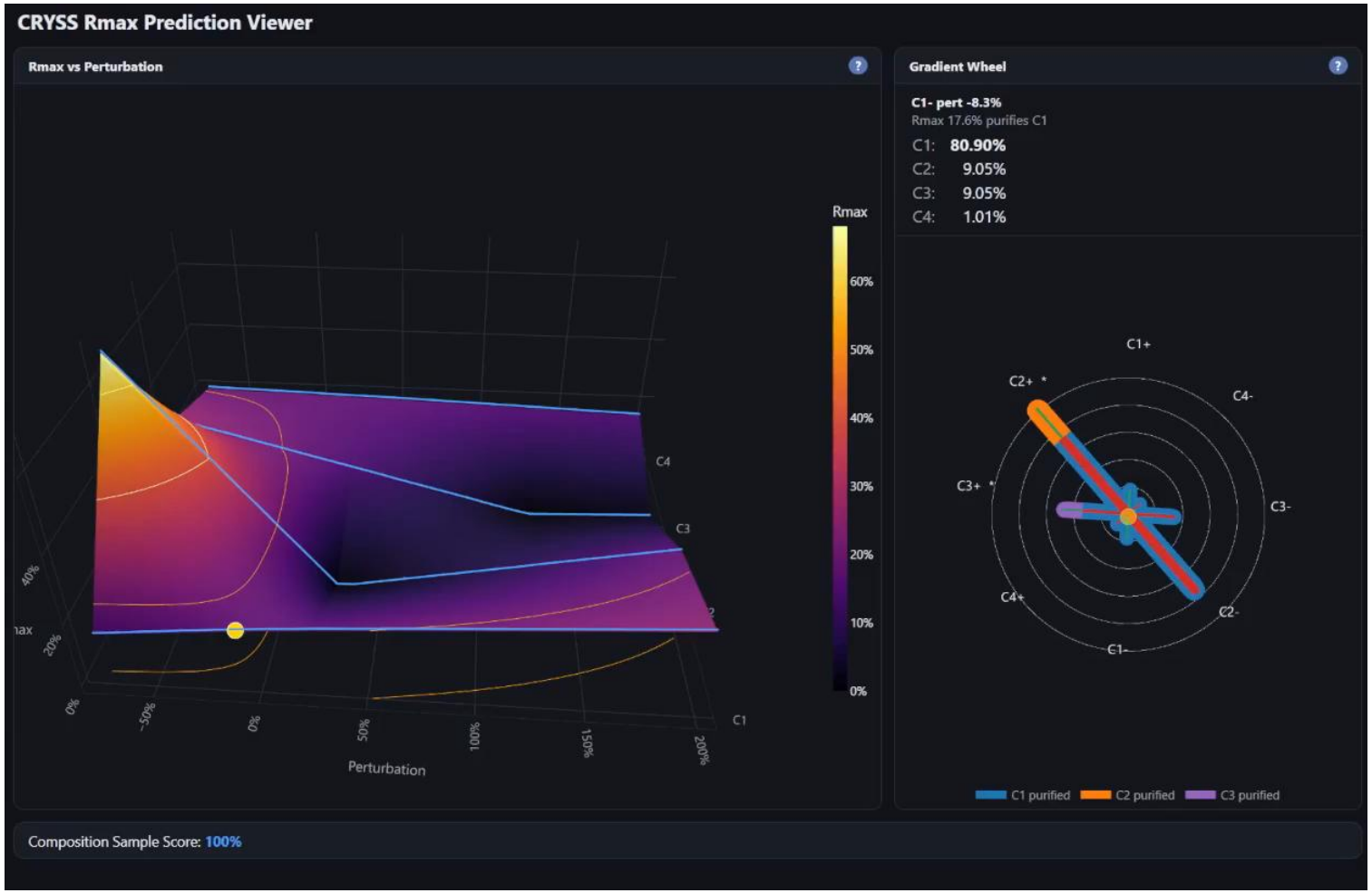


Well Mapped System



- If Recovery slice 1 captures all components in the solid
- If there are enough Recovery slices to cover ~ 60+% in the eutectic domain (100% - R_{max})
- The system will be well characterized and the fitted parameters will be much more than predictive for R_{max}
- They will have accurately encoded the underlying PhysChem relationships (at the equilibrium separation temperature in the given solvent system)
- CRYSS is built on a mechanistic model, which is fundamentally different that empirical response surface studies:-
  - A mechanistic model can be extrapolated
  - An empirical model can only be interpolated

CRYSS-T: Topography Mapping



- R_{max} surface developed by extending the initial composition space
- Gradient radar encoding sign, magnitude and discontinuities
- Can interrogate any starting composition and avoid identify regions of instability

Summary



- Experimental results confirm the eutectic cascade at the heart of CRYSS
- Process Screening:
  - CRYSS can be used to screen solvents and salt former conditions with minimal (1 – 2 solvent volumes) and meaningfully rank on the basis of Rmax, particular useful in early route scouting when transitioning from chromatography separation to more traditionally scalable crystallization mediated separations
- Process Optimization:
  - Multi-solvent volume slices allows creation of a predictive CRYSS-T model where high-risk QA/QC regions can be identified *in silico* and guide confirmatory experimentation before scale-up or validation

Acknowledgements



- Eric Damen
- Simone Darphorn-Hooijschuur
- Francois Gilardoni
- Ben McKay
- Lisa McQueen (née Agocs)
- Erik-Jan Ras
- Goran Verspui